

TBILISI, Georgia

WMO: 375490

Lat: 41.7500N

Lon: 44.7694E

Elev: 427

StdP: 96.30

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.7	(c) -3.1	(d) -9.8	(e) 1.7	(f) -1.1	(g) -8.1	(h) 2.0	(i) 0.3	(j) 19.6	(k) 4.5	(l) 17.3	(m) 4.8	(n) 2.9	(o) 320	(p) 0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.1	(c) 35.2	(d) 22.8	(e) 33.8	(f) 22.4	(g) 32.3	(h) 22.0	(i) 24.7	(j) 32.5	(k) 23.8	(l) 31.4	(m) 22.8	(n) 30.2	(o) 4.8	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 22.4	(b) 18.0	(c) 29.6	(d) 21.3	(e) 16.8	(f) 28.4	(g) 20.4	(h) 15.9	(i) 27.4	(j) 77.8	(k) 32.6	(l) 73.7	(m) 31.5	(n) 69.9	(o) 30.0	(p) 33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	16.2	14.0	-7.1	37.6	2.2	1.4	-8.7	38.6	-10.0	39.4	-11.2	40.2	-12.8	41.3
			-7.8	26.4	2.1	2.2	-9.3	28.0	-10.6	29.3	-11.7	30.6	-13.3	32.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	3.4	4.8	8.2	13.0	18.3	23.0	25.7	26.0	21.2	15.4	8.7	4.5		
	DBStd	8.70	2.77	3.32	3.42	3.81	3.19	2.98	2.55	2.86	3.18	3.45	3.46	3.30		
	HDD10.0	682	206	148	77	13	1	0	0	0	0	3	63	171		
	HDD18.3	2194	464	379	314	164	40	3	0	0	11	101	289	428		
	CDD10.0	2290	0	2	21	103	258	389	488	497	336	171	24	2		
	CDD18.3	760	0	0	0	4	38	142	230	238	97	11	0	0		
	CDH23.3	6922	0	0	2	54	282	1211	2239	2391	682	61	1	0		
	CDH26.7	2952	0	0	0	9	57	473	1036	1154	217	6	0	0		
(14) Wind	WSAvg	5.2	4.5	5.8	5.9	5.4	5.3	6.4	5.6	5.4	5.0	4.7	4.4	4.6		
(15) Precipitation	PrecAvg	583	24	26	38	72	87	86	46	43	44	54	38	25		
	PrecMax	800	54	55	85	123	204	169	86	96	114	146	91	91		
	PrecMin	472	3	7	6	25	30	16	5	4	3	9	2	2		
	PrecStd	91	12	12	21	29	34	39	21	25	29	31	24	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	17.1	22.0	27.8	30.6	35.0	36.8	37.4	33.3	27.2	20.2	15.5		
		MCWB	8.0	10.0	13.8	17.4	20.3	22.1	23.9	23.3	22.0	18.5	13.3	9.5		
	2%	DB	11.1	14.4	18.8	24.8	28.1	32.8	35.0	35.6	31.1	25.0	18.0	12.5		
		MCWB	6.7	8.7	11.8	16.3	18.9	21.4	22.9	22.9	21.0	17.4	12.2	8.0		
	5%	DB	9.2	12.1	16.2	22.2	26.4	31.2	33.6	34.2	29.1	23.0	16.0	10.7		
		MCWB	5.5	7.4	10.2	14.9	18.5	21.1	22.6	22.4	20.0	16.7	11.3	7.0		
	10%	DB	7.8	10.1	14.1	20.0	24.6	29.6	32.0	32.7	27.3	21.0	14.0	9.1		
		MCWB	4.6	6.1	9.1	13.7	17.6	20.6	22.3	22.2	19.4	15.7	10.4	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	11.5	14.6	18.5	22.1	24.2	25.8	26.5	24.1	20.0	14.6	10.3		
		MCDB	12.8	15.8	20.7	25.8	28.1	31.7	34.2	34.0	31.0	25.3	18.9	13.9		
	2%	WB	7.1	9.4	12.3	16.9	20.0	22.8	24.6	25.0	22.2	18.5	13.1	8.7		
		MCDB	10.3	13.4	17.4	23.4	26.1	30.5	32.9	32.6	28.7	23.3	16.2	11.5		
	5%	WB	5.9	7.7	10.9	15.5	19.0	21.9	23.7	23.9	21.2	17.4	11.9	7.5		
		MCDB	8.6	11.4	15.4	21.3	25.0	29.1	31.6	31.3	27.4	21.6	14.7	9.9		
	10%	WB	4.9	6.4	9.6	14.1	18.1	21.1	22.8	22.9	20.1	16.4	10.8	6.6		
		MCDB	7.3	9.5	13.3	18.9	23.7	28.1	30.1	30.3	25.6	20.0	13.5	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.2	9.3	10.0	10.2	10.9	10.8	11.1	10.3	8.9	8.5	7.2		
		MCDBR	10.2	12.5	13.6	14.3	13.2	13.5	13.3	13.7	13.1	12.1	11.9	9.7		
	5% WB	MCWBR	6.9	7.8	8.1	7.5	6.6	5.5	5.5	5.5	5.6	6.1	7.1	6.1		
		MCWBR	9.5	11.4	12.2	13.2	11.9	12.3	12.0	12.6	11.9	11.0	9.9	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.369	0.416	0.481	0.449	0.439	0.481	0.476	0.424	0.402	0.352	0.330		
	taud		2.319	2.204	2.115	1.978	2.145	2.232	2.130	2.136	2.262	2.259	2.354	2.344		
	Ebn at Noon		800	819	821	792	832	839	799	789	806	773	770	768		
	Edh at Noon		87	114	141	173	150	138	152	146	121	107	84	79		
(44) All-Sky Solar Radiation	RadAvg		1.87	2.77	3.47	4.32	5.22	6.09	5.97	5.49	4.30	2.87	2.02	1.61		
	RadStd		0.21	0.39	0.38	0.58	0.40	0.43	0.51	0.57	0.37	0.32	0.23	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.59	N/A	+1.19	+1.45	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page